



From Data & AI to Value Creation

(Re-)designing products, services, and human-machine collaboration

Andrea De Mauro

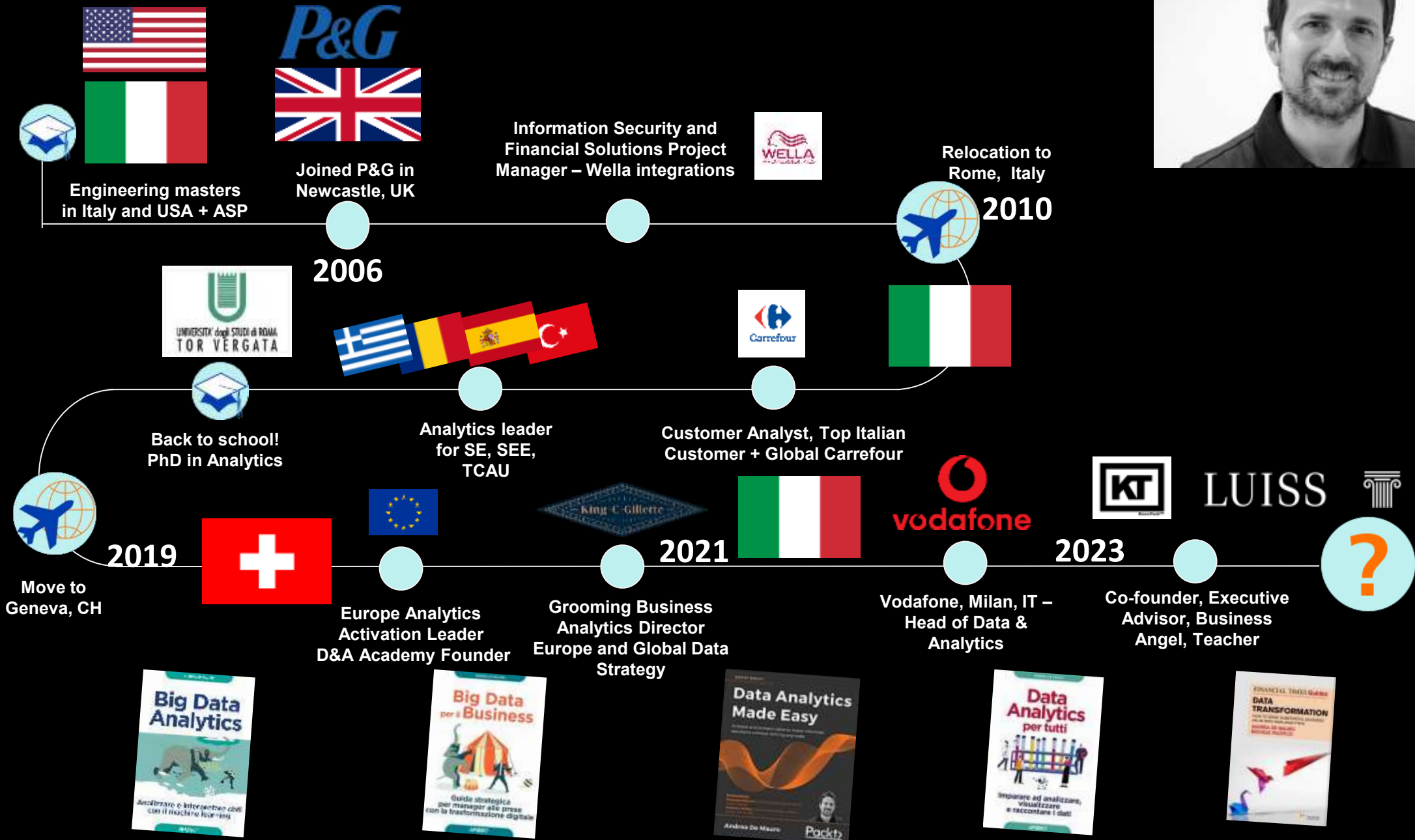
14/03/2026

An aerial photograph of a large crowd of people walking on a paved plaza. The plaza is marked with a complex network of thin, dark lines that form a series of interconnected triangles and polygons, creating a geometric pattern across the entire surface. The people are scattered throughout the scene, some walking alone, some in small groups, and some pushing strollers. The overall scene suggests a busy public space, possibly a park or a city square, where the geometric pattern might represent a network or data structure.

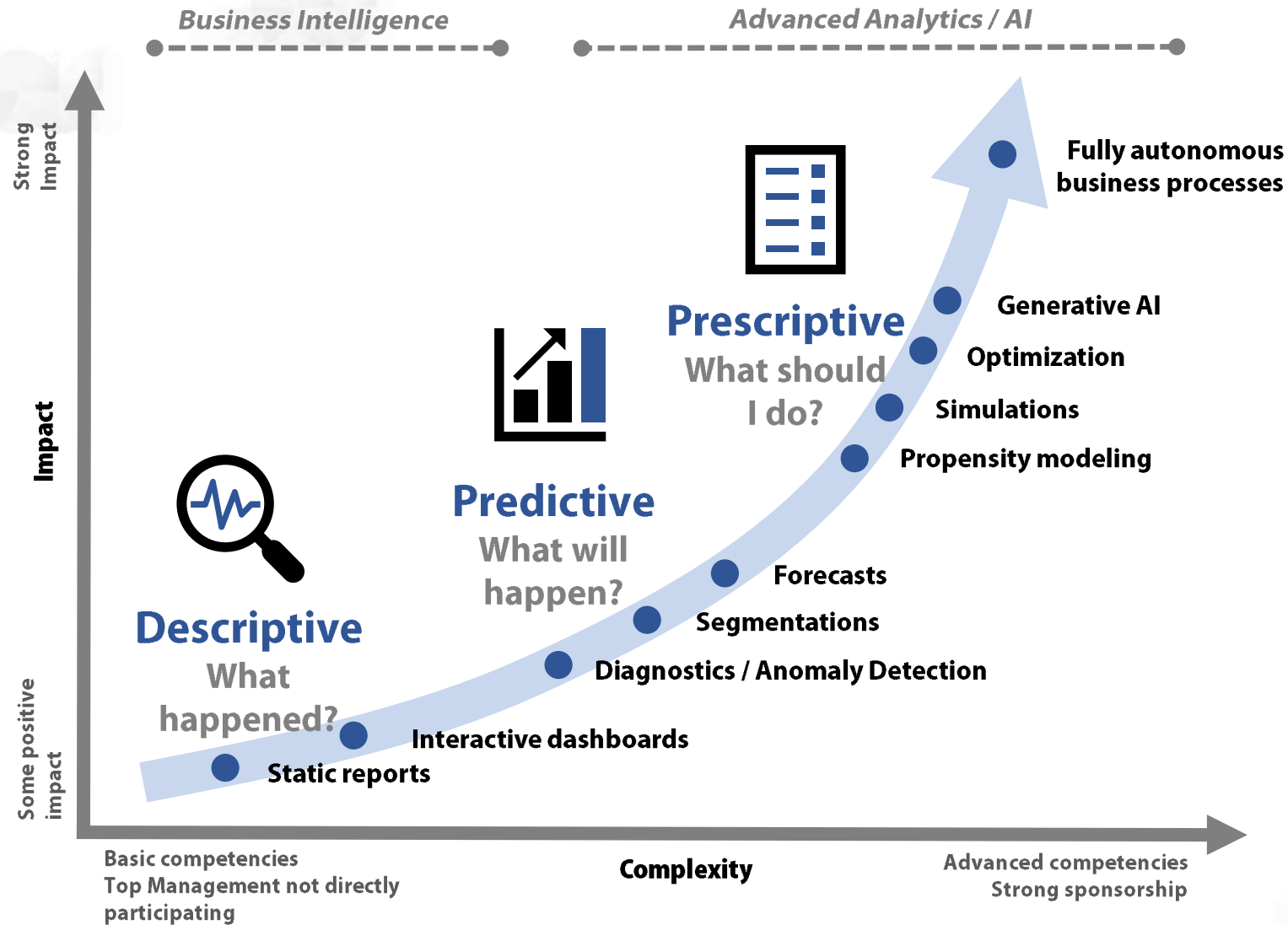
**Data analytics is pervasive in
modern business**

It involves each of you

Andrea De Mauro My path



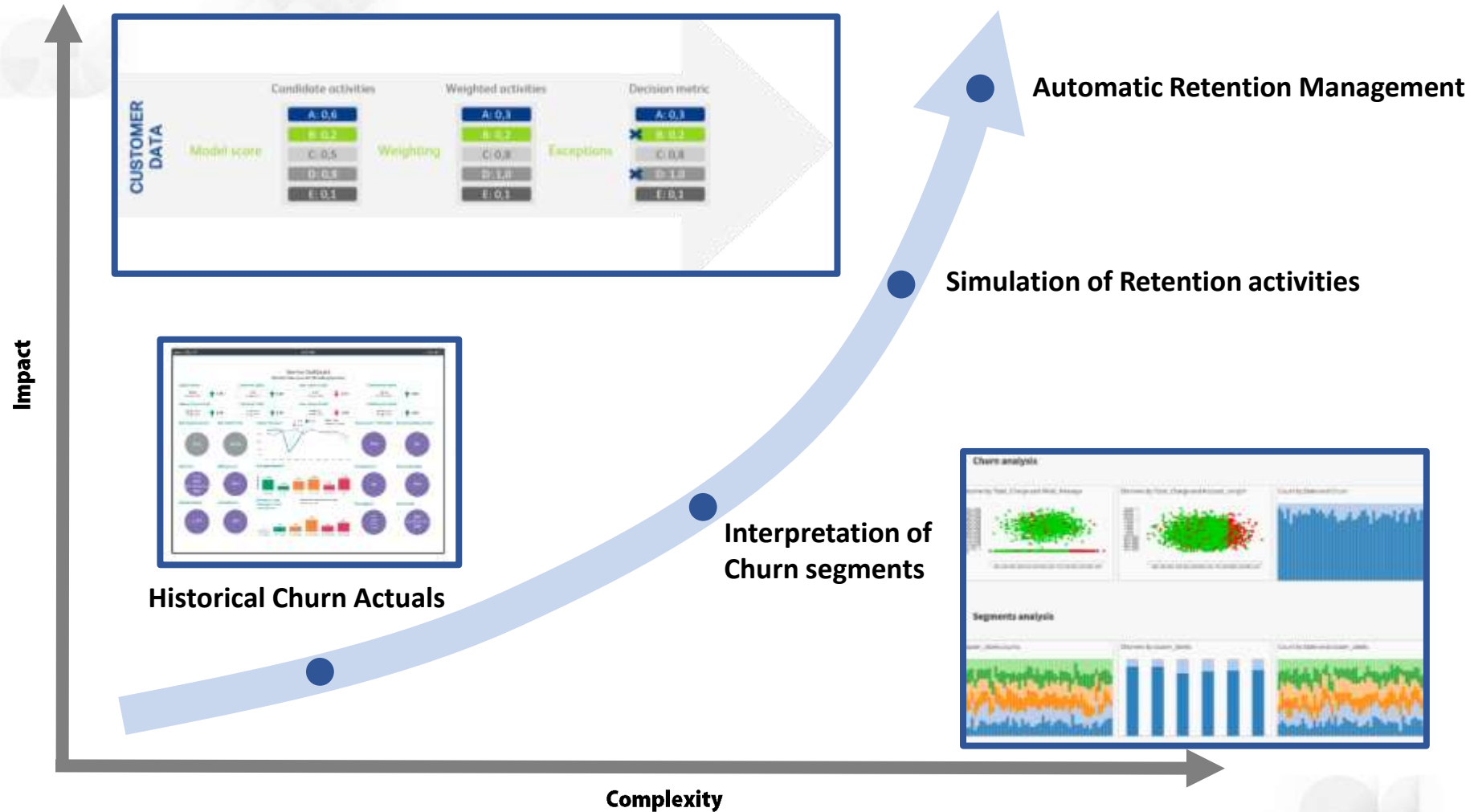
3 LEVELS OF DATA ANALYTICS



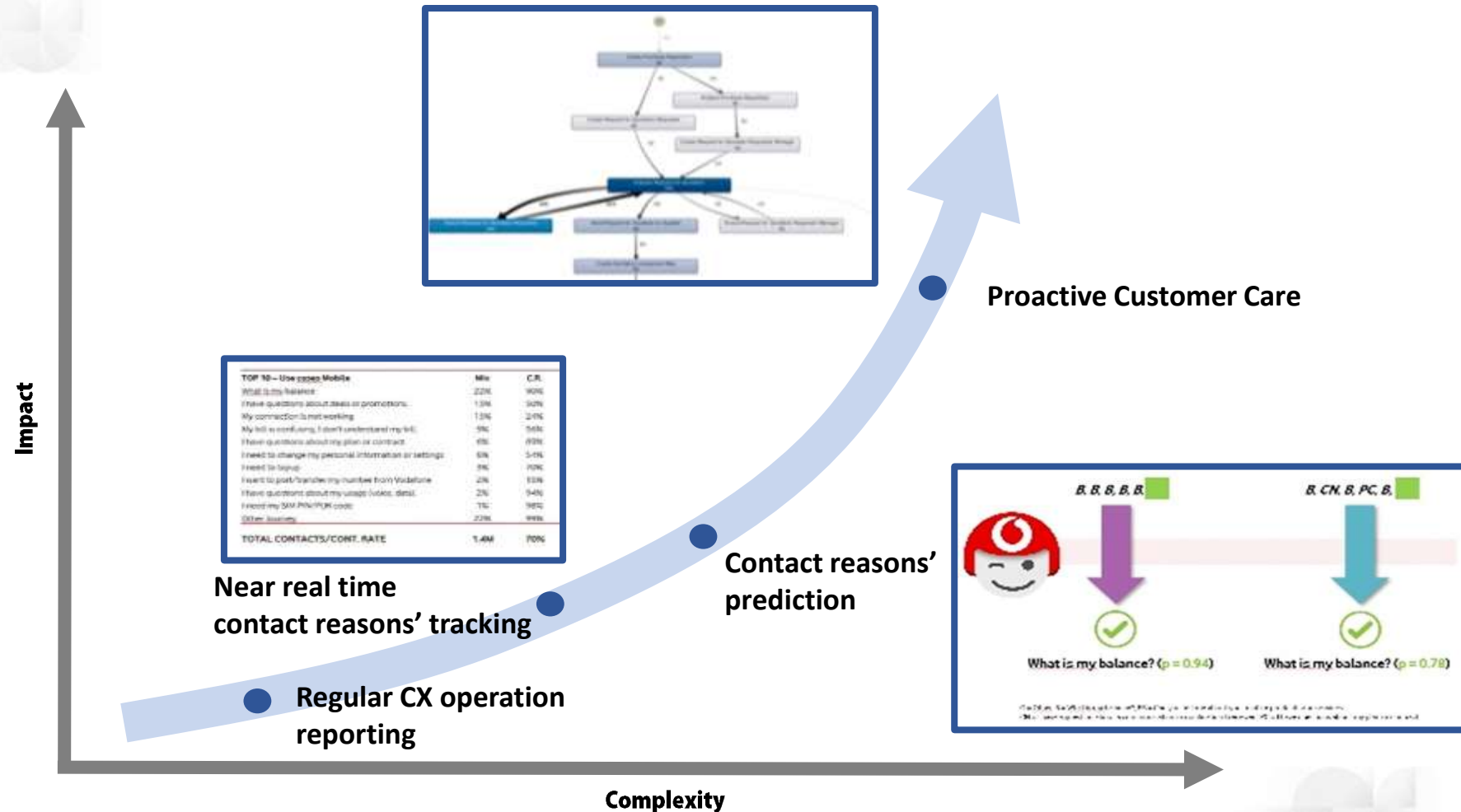
- 1) They are not alternatives. **They coexist.**
- 2) The same business need **requires them all.**

A FIRST TELCO EXAMPLE

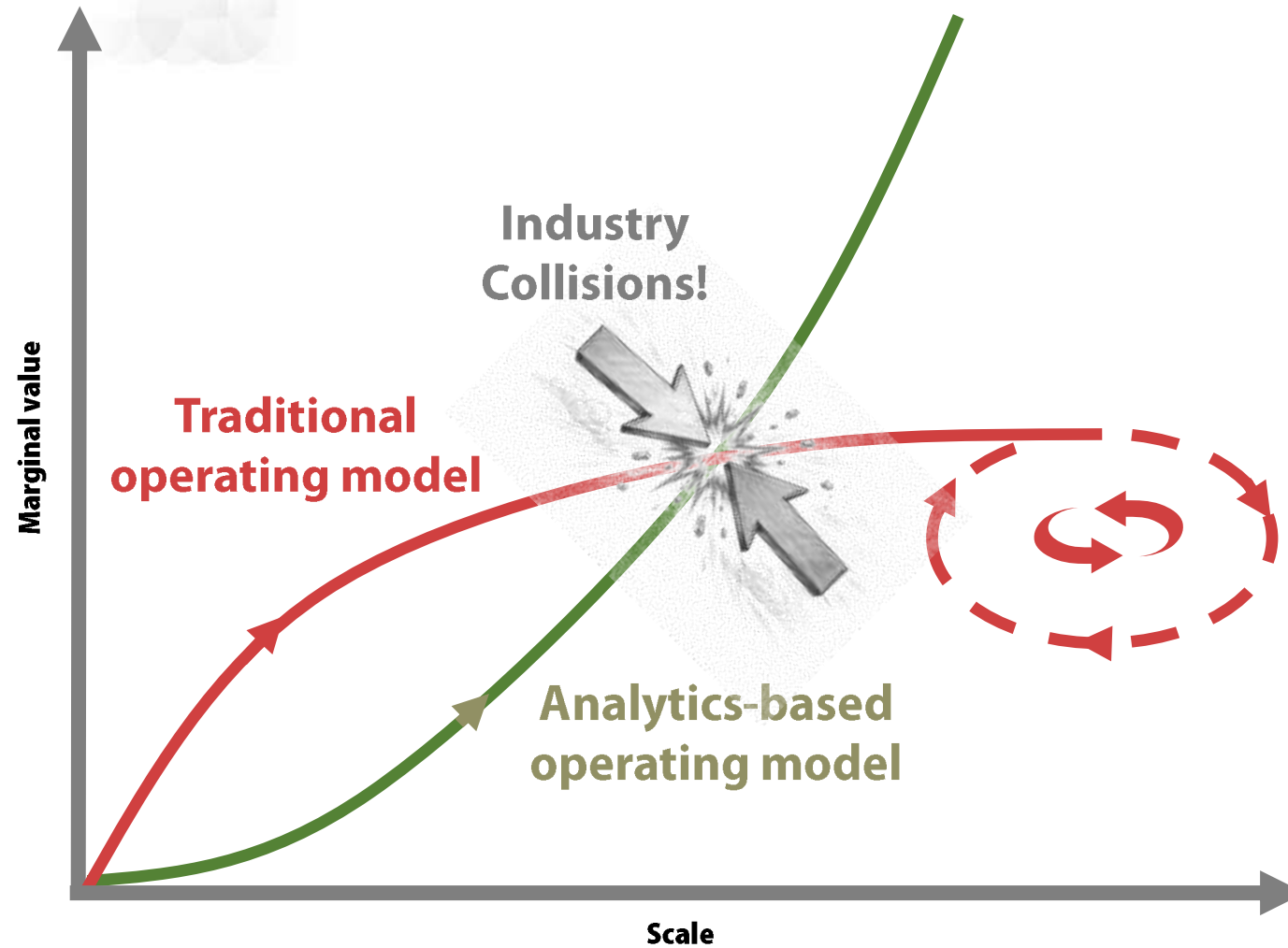
CHURN ANALYTICS



ANOTHER EXAMPLE BOOSTING CUSTOMER EXPERIENCE



CAN ANALYTICS TRANSFORM THE OPERATING MODEL?



Adapted from:

- Iansiti, M. and Lakhani, K. Competing in the age of AI, Harvard Business Press, 2020.
- De Mauro, A. Big Data per il business, Apogeo, 2020.

SEVERAL INDUSTRY COLLISIONS... AND MORE TO COME!

Instagram
facebook
airbnb
UBER
NETFLIX



Kodak

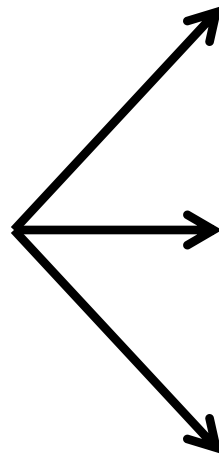
SAATCHI & SAATCHI

Hilton

TAXI

HOLLYWOOD

3 MAIN SOURCES OF VALUE FROM DATA ANALYTICS & AI

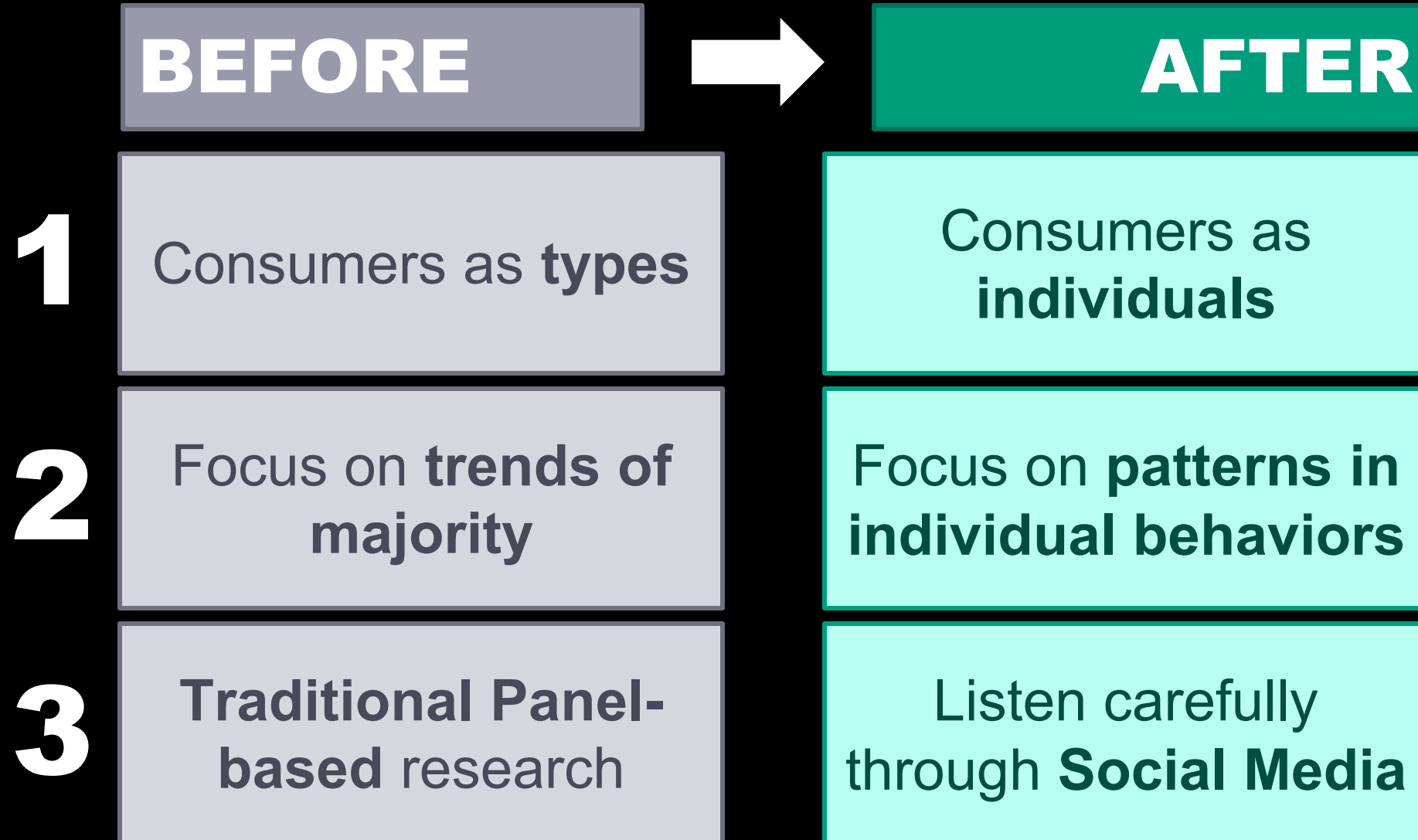


1. Cost reduction

**2. Better Decision
Making/Processes**

**3. Improved
Products/Services**

DATA-POWERED CONSUMER UNDERSTANDING



1. CONSUMERS AS “INDIVIDUALS”

Modern CRM systems recreate personalized interaction



2. FOCUS ON “PATTERNS”

amazon.it

Scegli per categoria - Ricerca Libri - Big Data

Libri Ricerca avanzata Tutti i generi Best seller Nostalgia Libri in inglese Libri in altre lingue Offerte Libri scolastici Libri universitari e professionali

Big data. Architettura, tecnologie e metodi per l'utilizzo di grandi basi di dati Copertina flessibile – 20 nov 2013
di **Alessandro Rezzani** (Autore)
★★★★★ = 3 recensioni clienti

• Visualizza tutte le 12 formate e le edizioni

Copertina flessibile
EUR 22,10
10 Nuovo da EUR 22,10

Vuoi riceverlo domani? Ordina entro **4 ore e 33 min** e scegli la spedizione 1 giorno. [Dettagli](#)

Ogni giorno nel mondo vengono creati miliardi di dati digitali. Questa mole di informazione proviene dal notevole incremento di dispositivi che automatizzano numerose operazioni - record delle transazioni di acquisto e segnali GPS dei cellulari, per esempio - e dal Web: foto, video, post, articoli e contenuti digitali generati e diffusi dagli utenti tramite i social media. L'elaborazione di questi "big data" richiede elevate capacità di calcolo, tecnologie e risorse che vanno ben al di là dei sistemi convenzionali di gestione e immagazzinamento dei dati. Il testo esplora il mondo dei "grandi dati" e ne offre una descrizione e classificazione, presentando le opportunità che possono derivare dal loro utilizzo. Descrive le soluzioni software e hardware dedicate, riservando ampio spazio alle implementazioni Open Source e alle principali offerte cloud. Si propone dunque come una guida approfondita agli strumenti e

• Leggi di più

Visualizza tutte le 2 immagini

Offerte speciali e promozioni

- Amazon Warehouse Deals, la nostra selezione di prodotti usati e ricondizionati in offerta. [Scopri di più](#)

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Nate Silver
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HYPER-ADAPTION

DATA COLLECTION

NETFLIX

230M+ 160+

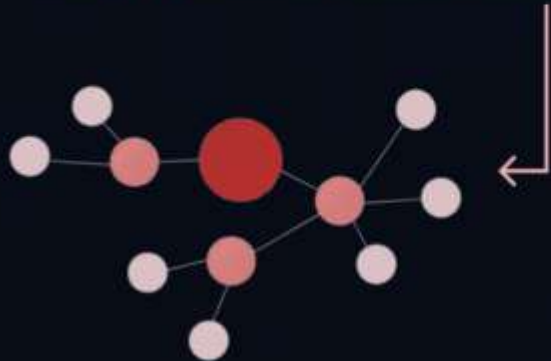
Engaging
Users

Countries

N



Netflix analyses millions of streams, using data like ratings and metadata across 1,300+ recommendation clusters



Artwork
Personalization

ADVANCED FEATURES

Contextual Bandits
Approach



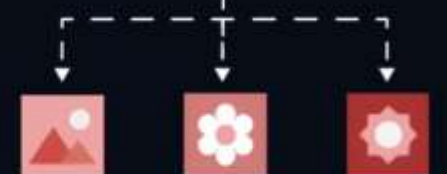
Netflix's search function tailors results to align with your viewing history and preferences

Search Personalization



Image selection algorithm personalizes thumbnails based on past interactions, increasing play clicks

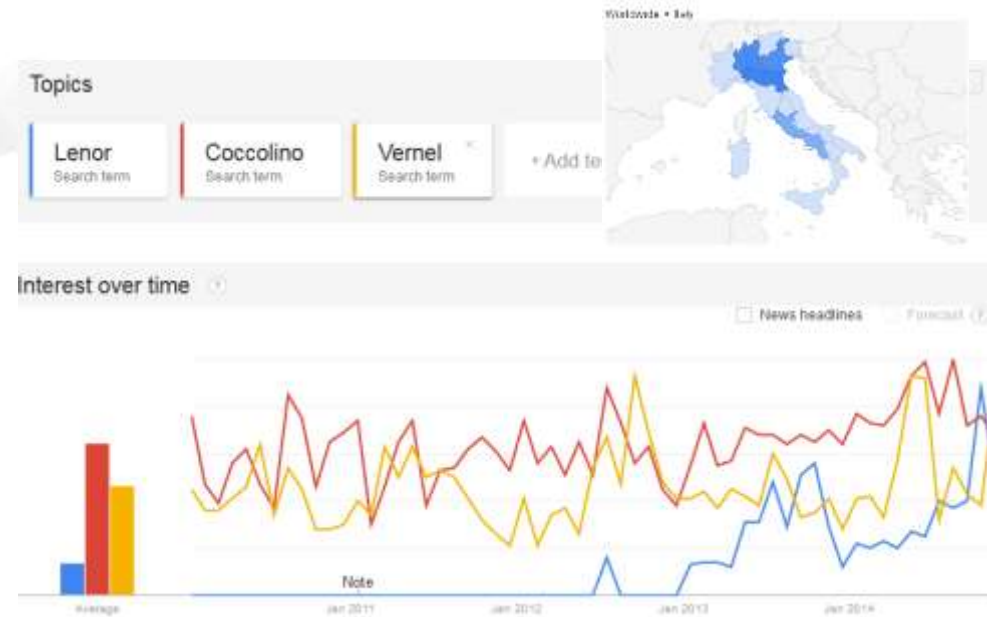
Click



Real-time adaptation to user preferences ensures relevant content, reducing outdated recommendations

3. LISTEN CAREFULLY

- Unprecedented sources
- Google Search logs
- Consumer comments
- Social media (geotagging)
- AI agents focus groups



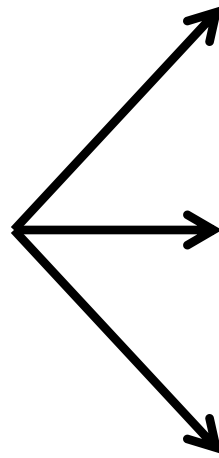
trends.google.com

The figure shows a customer review page for an Oral-B electric toothbrush. The title is 'Opiniones de clientes' and the product is 'Oral-B - Cepillo de dientes recargable Vitality Precision Clean, temporizador integrado'. It shows 249 reviews with a star rating breakdown: 3 stars (150), 4 stars (76), 2 stars (10), 1 star (1), and 0 stars (2). The average rating is 4.5 stars. A section titled 'La opinión positiva más útil' (Most useful positive opinion) features a 5-star review from Eduardo G. de Andres, dated February 7, 2012. The review praises the quality of the brush and its charging base. A 'Ver más opiniones de 5 estrellas, 4 estrellas' link is at the bottom.

The figure is a screenshot of the Brandwatch social media monitoring dashboard. It displays three columns of data for different brands. The first column shows a hand holding a smartphone. The second column shows a person in a racing suit. The third column shows a hand holding a blue cap with 'NIKE' written on it. Each column has a table of metrics: 'Mention volume', 'Distinct sources', 'Aggregate followers', and 'Latest activity'. The data for the first column is: Mention volume 1207, Distinct sources 51, Aggregate followers 986587, Latest activity 7 Jun 2017. The data for the second column is: Mention volume 19, Distinct sources 1, Aggregate followers 32439, Latest activity 13 Jun 2017. The data for the third column is: Mention volume 329, Distinct sources 1, Aggregate followers 125243, Latest activity 3 Jun 2017.

brandwatch.com

3 MAIN SOURCES OF VALUE FROM DATA ANALYTICS & AI



1. Cost reduction

**2. Better Decision
Making/Processes**

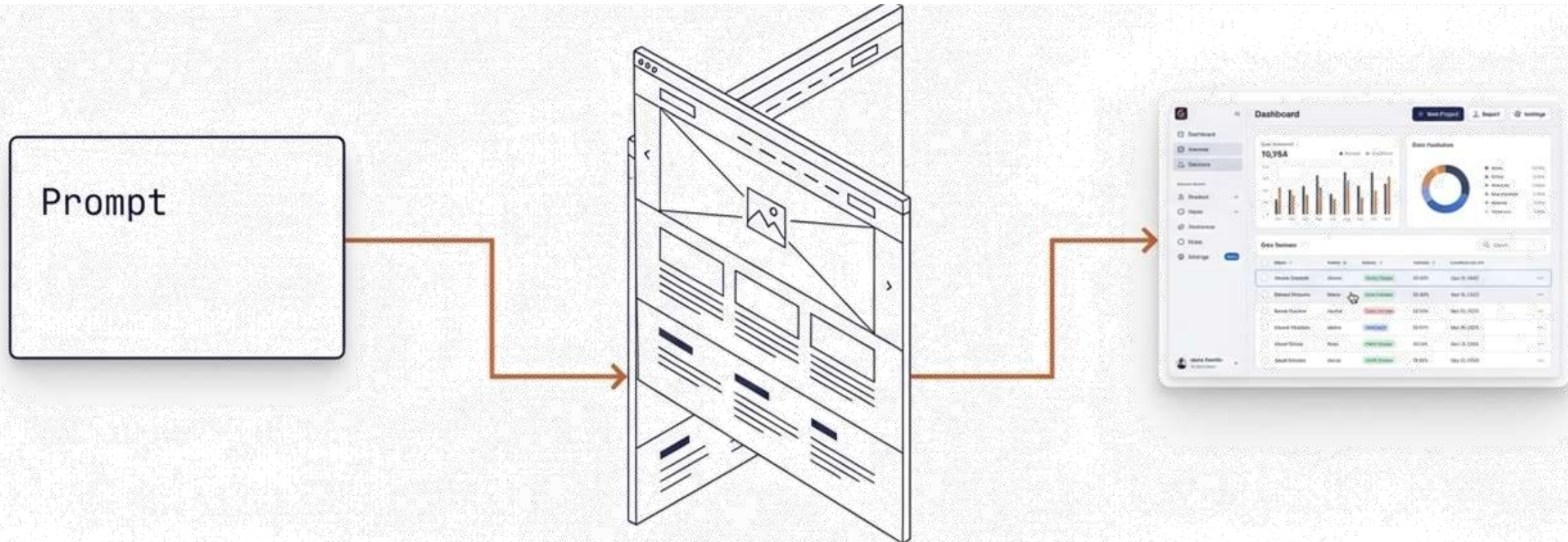
**3. Improved
Products/Services**

WHAT ABOUT IMPROVING THE **PROCESS** OF PRODUCTS' DESIGN?

CAN AI GENERATE?



RAPID PRODUCT PROTOTYPING WITH AI VIBECODING

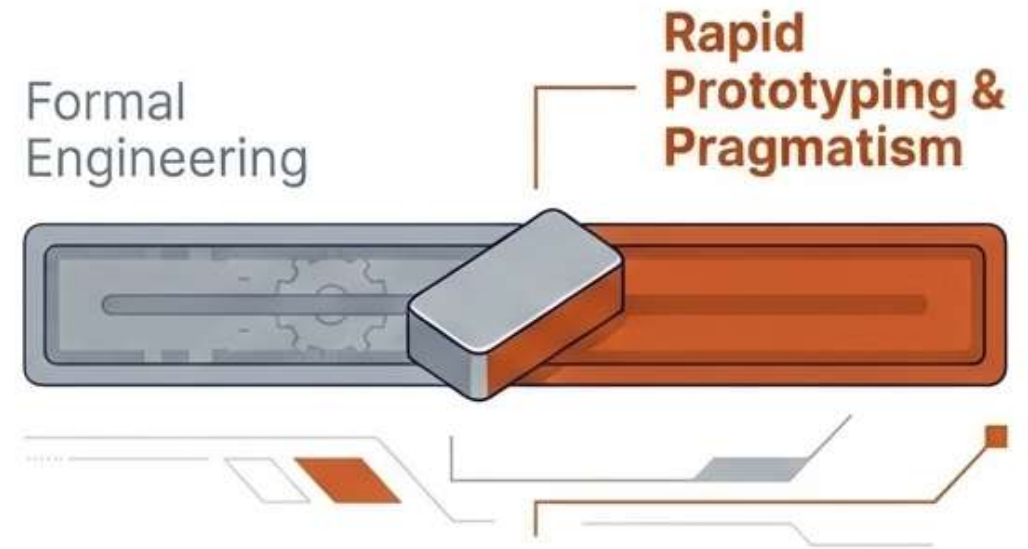


RAPID PRODUCT PROTOTYPING WITH **AI VIBECODING**

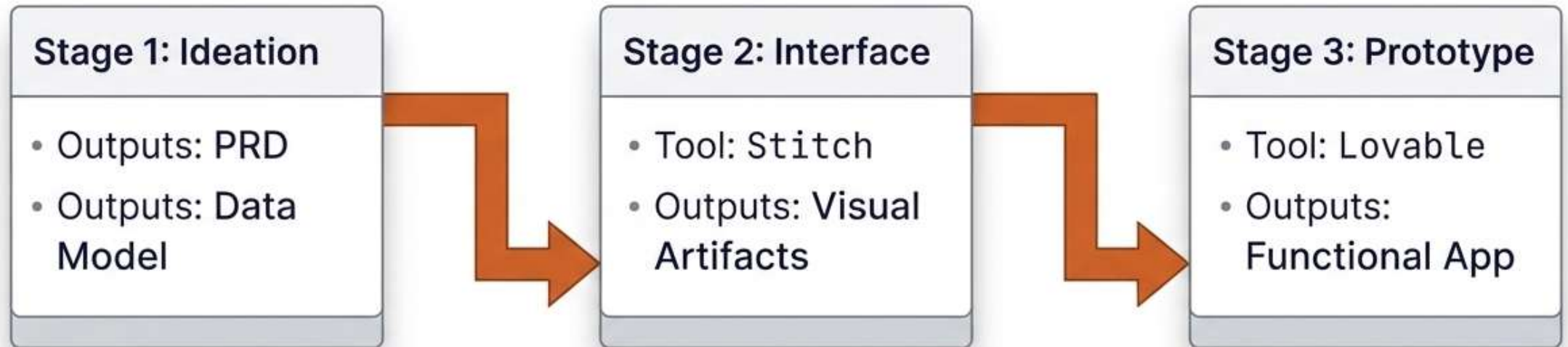
The Genesis of Vibe Coding

A software development methodology grounded in rapid experimentation, utilizing no-code and low-code tools without requiring formal engineering expertise.

Origin: 2025 | Coined by A. Karpathy



THE 3-STEP VIBECODING PIPELINE



STEP 1 - PRD



Generate a concise AI-ready PRD in markdown format, around 700–900 words, for a student-focused digital product called "...".

Follow exactly these 4 sections:

- 1. What to create*
- 2. Intended audience*
- 3. Rationale*
- 4. Operational framework*

Product concept:

...

Requirements for the PRD:

- Keep it practical and concrete*
- Focus on an MVP that could realistically be prototyped quickly*
- Do not include advanced AI features*
- Make it suitable for design and prototyping tools*
- Include the main user problem, the core features, and the expected user flow*
- Mention what data entities are needed for the product*
- Include constraints and assumptions*
- Include a short list of success criteria*

STEP 2 – UX SKETCH

*Based on the following PRD,
generate a visual product
concept and UX structure for a
web app called “...”.*

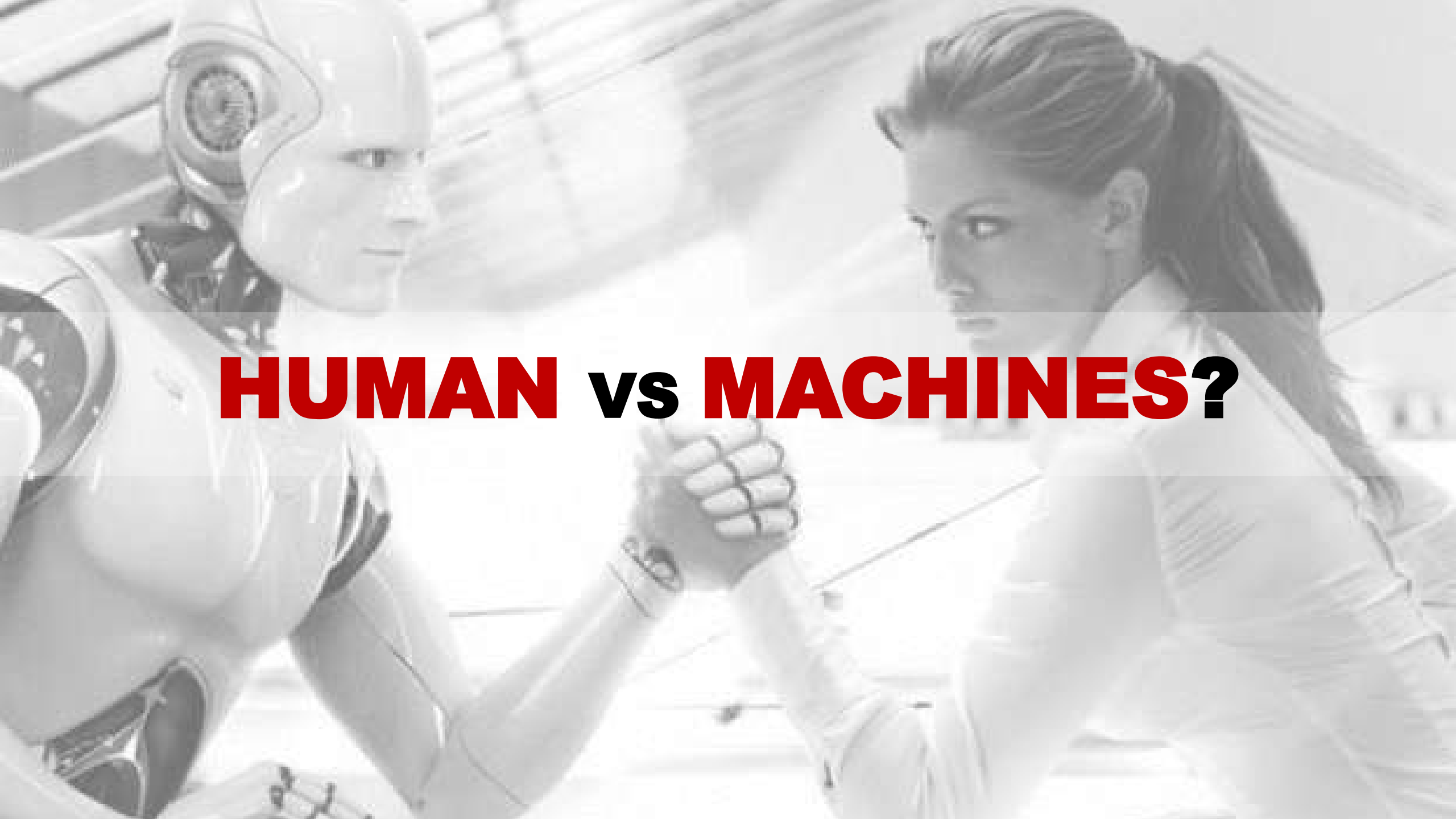
...



STEP 3 – LOVABLE

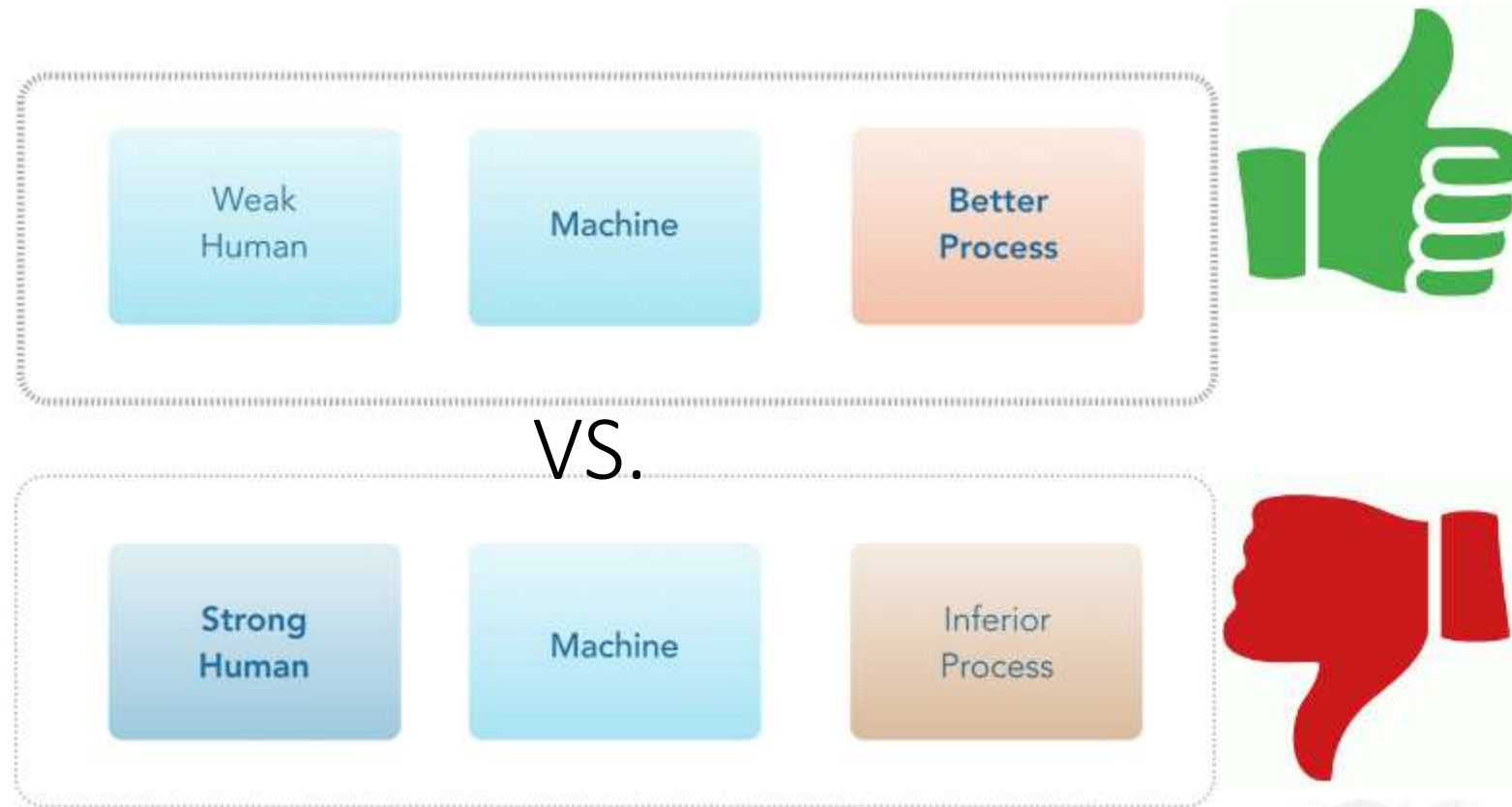
Build a responsive MVP web app
called... *based on ...*



A grayscale image depicting a tense moment between a humanoid robot and a woman. The robot, on the left, has a sleek, metallic body with visible joints and a human-like face. It is reaching out with its right arm, firmly grasping the woman's right arm. The woman, on the right, has long dark hair tied back and is wearing a light-colored, long-sleeved top. She is looking intently at the robot with a determined expression. The background is a bright, slightly out-of-focus indoor space with architectural lines. Overlaid in the center of the image is the text "HUMAN vs MACHINES?" in a bold, sans-serif font. "HUMAN" and "MACHINES?" are in red, while "vs" is in black.

HUMAN vs MACHINES?

KASPAROV'S LAW



Summing up...

What can we do with all of this?

1. We can create **real value** from data & algorithms leveraging the **three levels of analytics** and the **data-to-value paths**!
2. We can **design products with data** in mind!
3. Analytics is an unmissable key to **understanding consumers**!
4. The “secret” is in the human/machine quality of **collaboration**!
5. Key advice: **stay curious, learn, and experiment**

Thank you!

Stay in touch?

